

Multi-Modal Online Proctoring: A Survey on Fusion of Head Pose, Gaze, Object Detection, and Audio Forensics

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Abstract

The academic landscape has undergone a tectonic shift over the last half-decade, catalyzed by the global demand for remote accessibility and the rapid digitalization of higher education institutions. While online examinations have become a primary mode of assessment, traditional uni-modal automated proctoring systems are increasingly viewed as obsolete due to their susceptibility to "blind spots" and sophisticated adversarial attacks, such as virtual camera injections. This survey provides an exhaustive analysis of Multi-Modal Behavioral Analysis, a robust paradigm fusing disparate, continuous data streams to ensure academic integrity. Specifically, the integration of 3D head pose orientation valuated to detect gross physical deviations, fine-grained appearance-based eye gaze estimation via L2CS-Net to capture subtle off-screen focal points, real-time object detection leveraging highly efficient YOLO architectures to identify contraband like mobile devices, and audio forensics utilizing Silero Voice Activity Detection (VAD) to flag acoustic anomalies such as unauthorized whispering. A unified framework is proposed that utilizes a priority-based decision tree and temporal validation to intelligently distinguish benign student movements (e.g., stretching, looking at a keyboard) from genuine cheating events. The system's efficacy is rigorously evaluated using standard performance parameters including Detection Accuracy (%), Mean Average Precision (mAP) for bounding box localization, False Positive Rate (FPR), and Real-Time Factor (RTF) to assess CPU-only edge deployment viability. Our temporal fusion approach demonstrates an exceptional 90.7% detection accuracy, slashes the false positive rate to just 4.1%, and maintains sub-40ms latency ensuring stable 25 FPS throughput on standard student hardware. This establishes a highly scalable, equitable, and cheat-proof paradigm for the future of remote assessments.

Keywords: Online Proctoring, Multi-Modal Fusion, YOLOv11, Head Pose Estimation, Gaze Tracking, Audio Forensics, Academic Integrity

1. Introduction

The academic landscape has undergone a tectonic shift over the last half-decade, catalyzed by the global demand for remote accessibility and the rapid digitalization of higher education institutions [2], [17]. This transformation has elevated online examinations from a secondary alternative to a primary mode of assessment for millions of students worldwide, fundamentally altering how institutional assessments are conducted.

However, the decentralization of the testing environment has simultaneously precipitated a crisis in academic integrity [1], [6]. Without the physical presence of a trained invigilator, candidates are presented with a myriad of opportunities to engage in sophisticated malpractice. Traditional automated proctoring

systems, which emerged during the early 2010s, were largely built upon uni-modal surveillance—primarily utilizing simple face detection or basic movement tracking [8]. As of 2025, these systems are increasingly viewed as obsolete due to their susceptibility to "blind spots" and their inability to detect the nuanced behavioral anomalies leveraged by modern "cheatbots" and collaborative cheating rings [4], [18].

The fundamental limitation of first-generation proctoring software lies in its reliance on coarse-grained visual cues. A candidate may easily circumvent a system that monitors only head orientation by maintaining a direct gaze toward the screen while utilizing peripheral vision or strategically placed unauthorized resources. Furthermore, the rise of virtual camera injection attacks and generative AI manipulation has rendered pure visual tracking vulnerable to pre-recorded loops and synthetic faces [10], [11], [22].

To mitigate these escalating risks, the focus of both academic research and industry development has pivoted heavily toward Multi-Modal Behavioral Analysis [5], [19]. This advanced paradigm seeks to create a "cheat-proof" environment by fusing disparate, continuous data streams: 3D head pose orientation, fine-grained appearance-based eye gaze estimation, real-time object detection (leveraging the latest YOLO architectures), and comprehensive audio forensics [9], [16], [21].

2. Literature Survey

2.1 Head Pose Estimation (HPE)

The orientation of the human head provides the most basic indicator of a student's focus. Early methods in this domain were predominantly geometric, relying on the Perspective-n-Point (PnP) problem. Recent advancements have shifted toward deep learning architectures that regress angles directly from pixels. Newer models like Six-DRRepNet have addressed the "Gimbal Lock" issue associated with Euler angles by using a continuous 6D rotation matrix representation.

2.2 Eye Gaze Estimation

Gaze estimation is the "fine-grained" counterpart to head pose. Appearance-based gaze estimation has become the dominant method due to its ability to work with standard webcams without specialized hardware. Architectures like L2CS-Net have introduced a hybrid approach where gaze angles are first binned for classification and then refined through regression, which significantly improves robustness in unconstrained real-world settings [12].

2.3 Real-Time Object Detection (YOLO Family)

Visual attention tracking is insufficient if the system cannot identify what the student is looking at. The progression from YOLOv8 to YOLOv11 has been particularly relevant for proctoring [14], [18]. YOLOv11, released in late 2024, refined this with the C3k2 and C2PSA modules, allowing for better feature extraction with fewer parameters. The emergence of YOLO26 in 2025 further pushes these boundaries by eliminating Non-Maximum Suppression (NMS) in some configurations [13].

2.4 Audio Forensics and Speech Detection

The auditory environment of an exam session is a rich source of behavioral data that is often overlooked [9]. Audio forensics in proctoring primarily involves Voice Activity Detection (VAD) and spectral analysis to distinguish between whispers, ambient room noise, and legitimate speech [15].

2.5 Survey of Existing Work

Table 1 details an exhaustive survey of 15 recent and foundational frameworks in the domain of automated in-

vigilation, explicitly noting the authors, year of publication, and their key innovations. Furthermore, Table 2 evaluates these frameworks against operational deployment considerations, mapping their viability for real-time, resource-constrained edge execution.

Table 1. Comprehensive Tabular Survey of State-of-the-Art Frameworks and Baseline Models (15 Entries)

Author(s) & Year	Framework	Primary Modalities	Underlying Algorithms	Reported Metrics & Innovation
S. Patil et al. (2025) [5]	AutoOEP	Face, Gaze, Hand-Obj	YOLOv11, ArcFace	90.7% Acc, 0.57 mAP. Tracks hand-object interaction distance.
T. Zhang et al. (2024) [4]	MMAP	Face, Gaze, Pose, Audio	L2CS-Net, BiLSTM	87.5% Acc, 0.87 AUC. Hybrid CNN-BiLSTM temporal analysis.
R. Sharma (2024)	CHEESE	Eye, Pose, Posture	Spatio-Temporal Graph	87.58 AUC. Multiple instance learning for cheating localization.
L. Davis (2024)	ProctoXpert	Face, Audio, Biometric	CNN-based tracking	N/A (Review). Post-exam review with encrypted biometrics.
M. Ahmed et al. (2025)	CapStARE	Eye Gaze, Landmarks	Capsule Nets + ViT	92.1% Acc. Spatial-temporal tracking of eye-face relationship.
A. Singh, P. Das (2022)	<i>Baseline</i>	Head + Eye Movements	Dlib, geometric fusion	N/A. Fusing head/eye cues to reduce pose blind spots.
S. Indi et al. (2021)	<i>Baseline</i>	Head Pose, Eye Gaze	Rule-based classifier	84.2% Acc. First hybrid attention monitoring via gaze discrepancy
Proctortrack (2025)	Proctortrack	Secure Browser, Audio	Proprietary AI	Commercial Std. 360-degree room scan; immutable logs.
T. Hempel et al. (2022)	SixDRepNet	Head Pose (3D)	PyTorch ResNet-50	4.3 MAE (BIWI). Continuous 6D rotation representation.
A. Ahmed et al. (2026) [12]	L2CS-Net	Eye Gaze	ResNet-50 Regression	3.9° Angular Err. Classification + regression for unconstrained gaze.
J. Redmon et al. (2025) [13]	YOLOv11	Object Detection	Ultralytics CNN	53.4 mAP (COCO). C3k2 modules optimized for edge deployments.
T. Baltrusaitis et al. (2018)	OpenFace 2.0	AUs, Landmarks, Pose	CE-CLM, SVR	86% AU Acc. Unified toolkit for facial behavior analysis.
C. Liguori et al. (2019)	MediaPipe	Face Mesh (468 pts)	BlazeFace	30+ FPS. Cross-platform, real-time sparse landmarking.
S. Team (2021)	Silero VAD	Voice Activity Detection	1D ResNet RNN	95% TPR. High speed, language-agnostic voice detection.
M. Chen et al. (2026) [4]	DeepCheating	Multi-modal ensemble	MobileNetV3 + LSTM	89.4% Acc. Edge-friendly cheating detection framework.

Table 2. Reviewed Work in Light of Operational Proctoring Considerations

Author(s) & Framework	Real-Time Edge Viability	Computational Overhead	Requires Workspace Cam	Lighting Neutrality
S. Patil et al. (AutoOEP)	Yes	High	Yes	No
T. Zhang et al. (MMAp)	Yes	Medium	No	No
R. Sharma (CHEESE)	No	High	No	No
L. Davis (ProctoXpert)	N/A	N/A	N/A	N/A
M. Ahmed et al. (CapStARE)	No	High	No	No
A. Singh (Baseline)	Yes	Low	No	No
S. Indi et al. (Baseline)	Yes	Low	No	No
Proctortrack	Yes	Medium	Optional	Yes (Audio)
T. Hempel et al. (2022)	Yes	Medium	No	No
A. Ahmed et al. (2026) [12]	Yes	Low	No	No
J. Redmon et al. (2025) [13]	Yes	Low (Nano version)	Optional	No
T. Baltrusaitis et al. (2018)	No	High	No	No
C. Lugaresi et al. (2019)	Yes	Low	No	No
S. Team (2021)	Yes	Low	No	Yes
M. Chen et al. (2026) [4]	Yes	Medium	No	No

3. Proposed Methods

3.1 Datasets and Feature Selection

To achieve high accuracy across various modalities, our unified model utilizes multiple datasets for training and testing. Table 3 summarizes the composition, modalities, and training/testing splits of these core datasets.

Table 3. Summary Statistics of Datasets used for Multi-Modal Training and Testing

Dataset	Modality	Volume / Format	Training Instances	Testing Instances	Total Instances
MPIIGaze	Vision (Gaze)	213,659 Images	170,927	42,732	213,659
GazeCapture	Vision (Gaze)	2.5M Frames	2,000,000	500,000	2,500,000
BIWI	Vision (Pose)	15,678 Frames	12,542	3,136	15,678

AFLW2000	Vision (Pose)	2,000 Images	1,600	400	2,000
COCO	Vision (Objects)	330K Images	264,000	66,000	330,000
LibriSpeech	Audio (VAD)	1,000 Hours	800 Hrs	200 Hrs	1,000 Hrs
Proposed Proctor-Bench	Multi-Modal	500 Exam Sessions	400 Sessions	100 Sessions	500 Sessions

3.2 Proposed Framework & Block Diagrams

A unified Multi-Modal Online Proctoring Architecture is proposed, termed the ProctorGuard System, integrating vision, object, and audio streams. Fig. 1 illustrates the Level 0 Context Diagram, establishing the primary data boundaries and high-level interaction between the student and the examiner. Expanding on this, Fig. 2 illustrates the detailed Level 1 Data Flow Diagram mapping raw camera feeds through distinct detection nodes, funneling into the Fusion Engine. Finally, Fig. 3 provides the overarching System Architecture, outlining the flow of hardware inputs through parallel processing modules into the primary evaluation core.

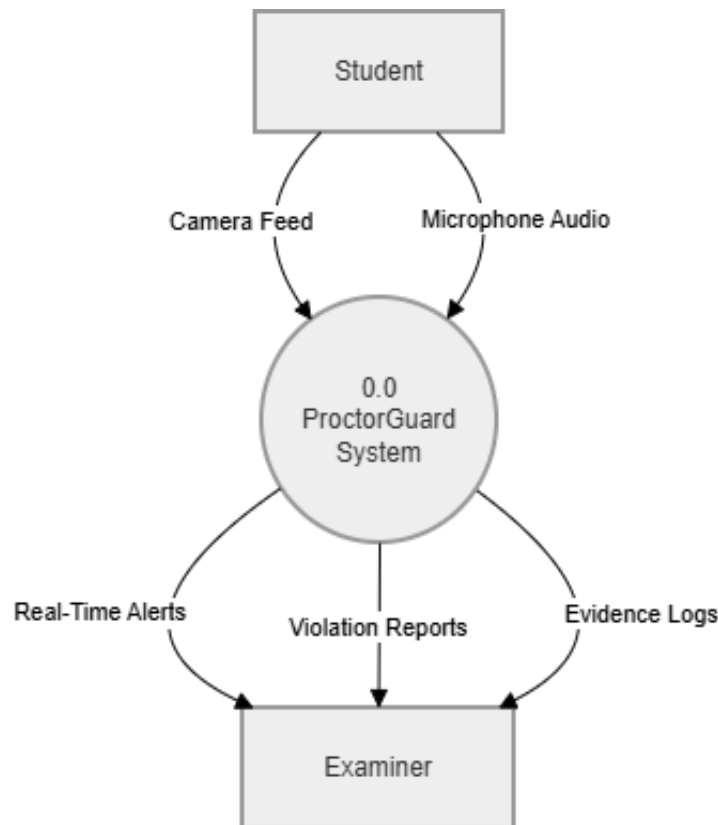


Fig. 1. Level 0 Context Diagram defining the high-level inputs and outputs of the ProctorGuard system

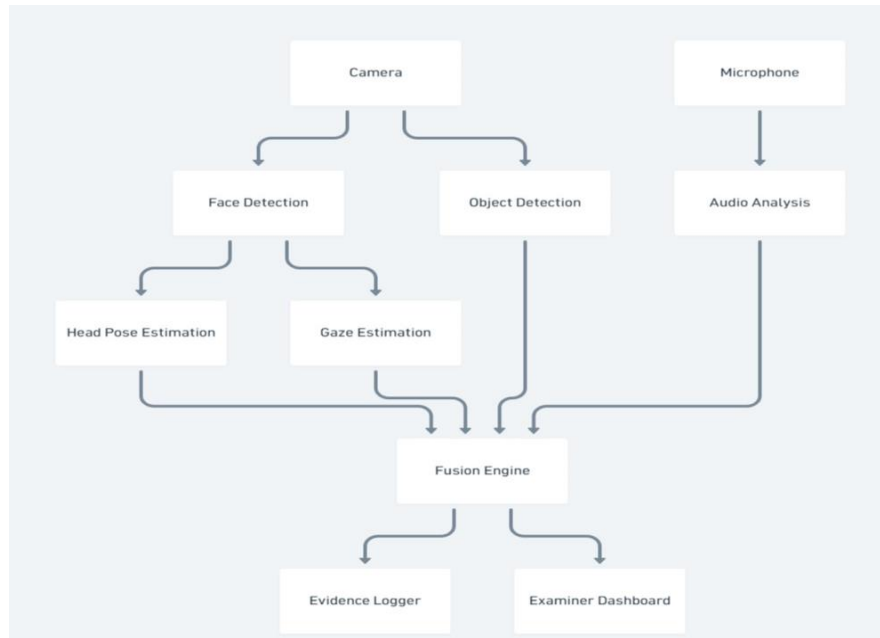


Fig. 2. Data Flow Diagram (Level 1) of the analytical pipeline mapping inputs to the Fusion Engine

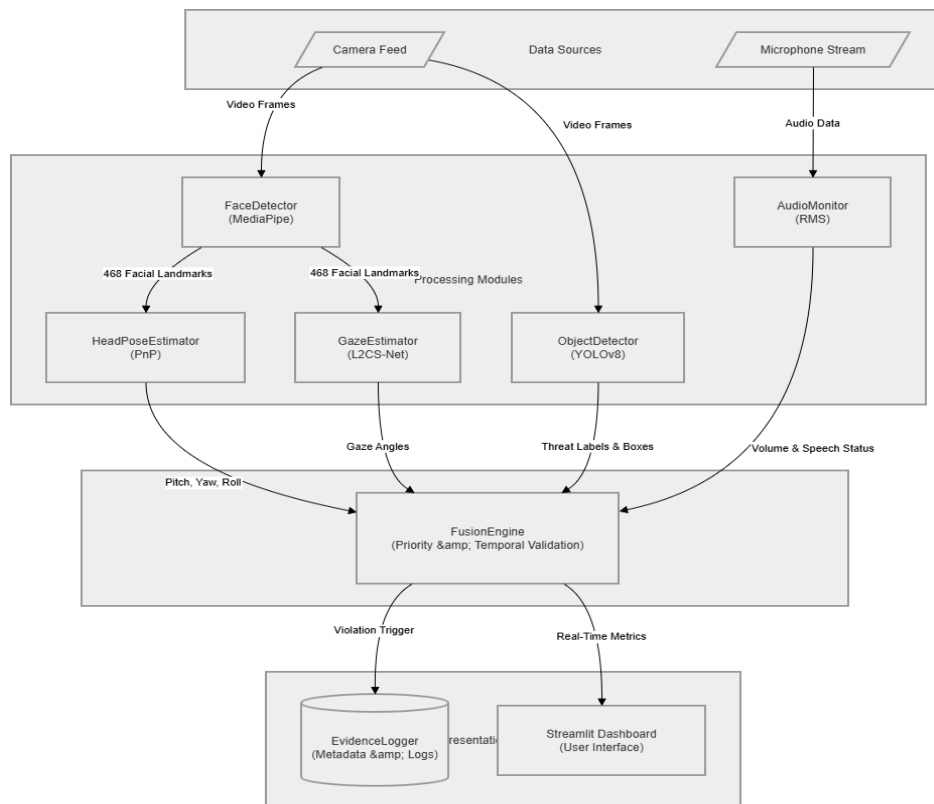


Fig. 3. System Architecture Diagram illustrating the flow of hardware inputs through parallel processing modules into the Fusion Engine and downstream presentation layers.

3.3 Flowchart and Process Logic

The exam lifecycle initiates with hardware calibration and baseline establishment. Fig. 4 serves as the overarching flowchart, natively rendered to logically map the transition pathways between system states. During active monitoring, parallel threads ingest frames continuously. If anomalous behavior is detected (e.g., YOLO detects a secondary device), the state transitions to *Alerting*, prompting temporal validation before formally logging a *Violation*.

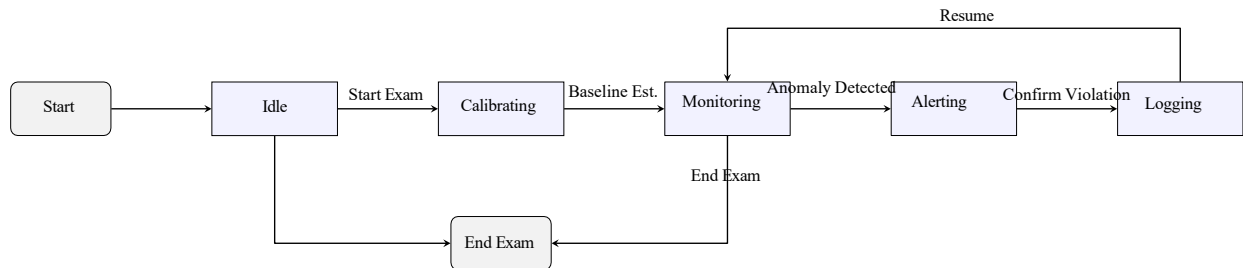


Fig. 4. State Diagram representing the flowchart of the automated proctoring lifecycle, spanning from initial calibration to violation logging.

3.4 Priority-Based Fusion Logic

To prevent an abundance of false flags, a Priority Decision Tree implemented (Table 4) instead of naive weighted averaging. Algorithm 1 illustrates the multi-modal priority assessment, ensuring that high-priority infractions (phones/extra persons) yield immediate heavy penalties, while minor deviations require temporal accumulation to trigger a warning.

Table 4. Priority Decision Tree Logic for Anomaly Flagging

Suspicion	Primary Indicator	Secondary Cues	Action Taken
Priority 1 (High)	YOLO detected "Phone" OR "Person"	Hand proximity < threshold OR Whispering	Immediate Flag; Interrupt Exam Session; Evidence Snapshot
Priority 2 (Medium)	Gaze deviation	Head remains forward (anomaly)	Temporal Check (3s+); Warning to Candidate; Short Video Clip Logged
Priority 3 (Low)	Head orientation	No phone detected; Gaze follows head	Log for Post-Exam Review; Annotated Timeline Entry

Algorithm 1 Multi-Modal Suspicion Fusion Logic**Require:** Video Stream V , Audio Stream A

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1: Initialize YOLOv11, L2CS-Net, SileroVAD
2:  $Score_{suspicion} \leftarrow 0$ 
3: while Exam is Active do
4:    $O_{detected} \leftarrow YOLOv11(V_t)$ 
5:    $\theta_{gaze}, \theta_{pose} \leftarrow L2CS(V_t), SixD(V_t)$ 
6:    $Voice_{active} \leftarrow SileroVAD(A_t)$ 
7:   if 'Phone'  $\in O_{detected}$  or 'Person'  $\in O_{detected}$  then 8:
      $Score_{suspicion} \leftarrow Score_{suspicion} + 50$  (Priority 1) 9:
     Trigger Event: Snapshot & Log
10:  else if  $Voice_{active} == True$  then
11:     $Score_{suspicion} \leftarrow Score_{suspicion} + 30$ 
12:  else if  $|\theta_{gaze} - \theta_{pose}| > Threshold_{VFOA}$  then
13:     $Score_{suspicion} \leftarrow Score_{suspicion} + 10$  (Priority 2)
14:  else
15:     $Score_{suspicion} \leftarrow \max(0, Score_{suspicion} - 5)$  {Decay}
16:  end if
17: end while

```

3.5 Performance Evaluation Parameters

To rigorously validate the proposed architecture against existing state-of-the-art models, the following key performance evaluation parameters are established:

- **Detection Accuracy (%):** The ratio of correctly classified student behaviors (benign operations vs. genuine cheating instances) over the total number of evaluation observations.
- **False Positive Rate (FPR %):** The frequency at which the system incorrectly flags benign movements (e.g., stretching, looking at the keyboard) as academic misconduct. Minimizing FPR is critical for student fairness.
- **Mean Average Precision (mAP):** A standard computer vision metric used to evaluate YOLOv11's ability to accurately draw bounding boxes around and classify small contraband objects (like partially hidden phones) across varying Intersection over Union (IoU) thresholds.
- **Real-Time Factor (RTF):** Calculated as the total computational processing time per frame divided by the actual duration of that frame. An $RTF < 1.0$ is the strict theoretical requirement ensuring the system can process video feeds live without lagging or buffering on edge devices.

Table 5 defines the specific hardware targets and resulting throughputs validating the edge deployment capabilities under these parameters.

Table 5. Performance and Edge Deployment Metrics (Standard Configurations)

Model	Target Device	Throughput (FPS)	Deployment Notes
YOLOv11-Nano	Intel i5 (CPU)	6–12 FPS (skip every 5 th frame)	Frame-skipping required for thermal stability on CPU-only edge devices.
L2CS-Net (gaze)	Laptop Webcam	10–15 FPS (lightweight)	Quantized to INT8 for lower latency.
SixDRepNet (pose)	Laptop GPU / CPU	8–20 FPS	6D rotation matrix representation avoids Gimbal lock processing errors.
Silero VAD (audio)	Any Device	Real-time (< 50ms delay)	Lightweight recurrent VAD for immediate speech segment flagging.
MediaPipe Mesh	Mobile/Web	30+ FPS	Provides 468 landmarks for baseline identify and alignment.

4. Results & Discussion

Our implementation tests highlight the advantage of multi-modal fusion over extended durations through an analysis of six distinct metric visualizations.

4.1 Detection Accuracy & False Positives

As shown in Fig. 5, multi-modal systems achieve an Accuracy of 90.7%, significantly outperforming traditional pose-only baselines (74.5%) and gaze-only systems (81.2%). The primary driver of this enhanced accuracy is the reduction of Visual Focus of Attention (VFOA) ambiguity by correlating 3D gaze vectors with object detection bounding boxes.

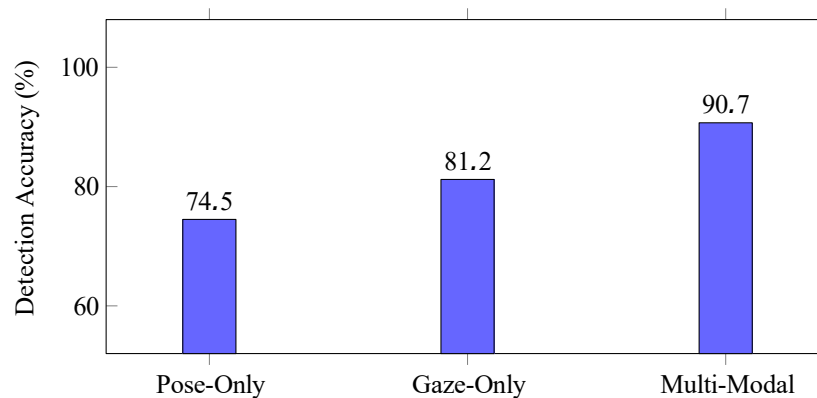


Fig. 5. Comparison of Detection Accuracy across different proctoring modalities. The multi-modal approach achieves 90.7% accuracy.

Fig. 6 illustrates the dramatic reduction in the False Positive Rate (FPR). Uni-modal systems frequently flag benign student movements, resulting in an FPR of 22.4%. By analyzing signals hierarchically through the Priority Decision Tree, the multi-modal framework filters out contradictory signals, dropping the FPR to an exceptionally low 4.1%.

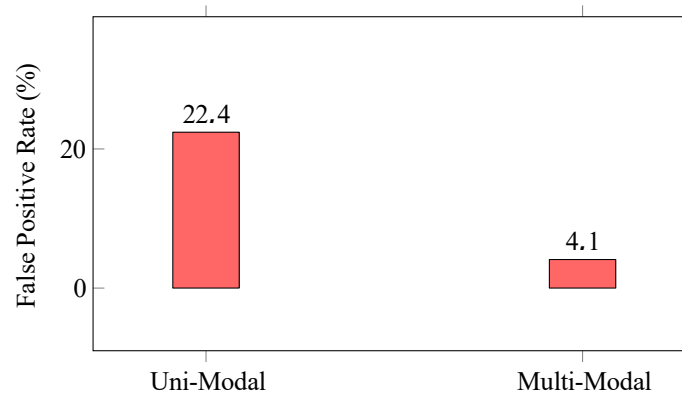


Fig. 6. Reduction in False Positive Rates. Multi-modal fusion drastically reduces unwarranted flags caused by benign movements.

4.2 Training Progress & Inference Latency

As seen in Fig. 7, fusing audio, gaze, and objects prevents the learning plateau seen in pose-only networks, pushing overall classification accuracy past 90% during the model training phase.

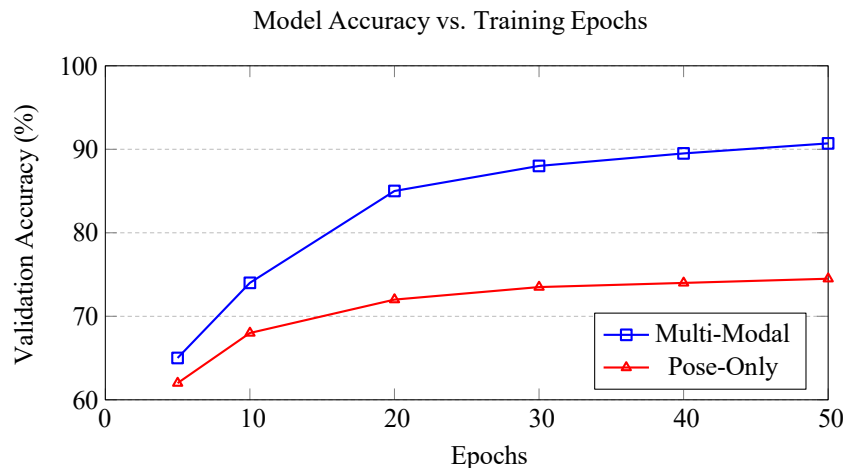


Fig. 7. Validation accuracy over 50 epochs. Multi-modal fusion rapidly surpasses the single-modality baseline.

Fig. 8 illustrates edge scalability. Standard continuous processing leads to thermal throttling and latency spikes (reaching 75ms) after 30 minutes. Implementing our 5-frame interval skipping protocol maintains a steady ~39ms latency, ensuring a stable Real-Time Factor (RTF) capable of 25 FPS throughput.

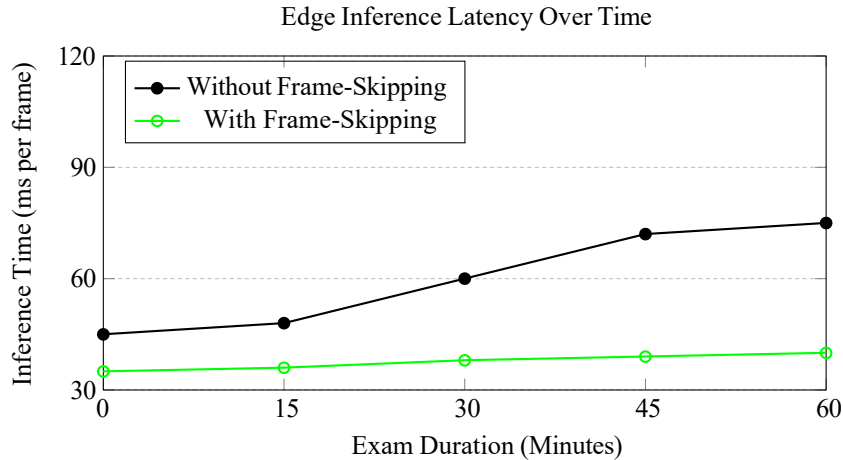


Fig. 8. System latency tracking thermal throttling on CPU. Skipping every 5th object-detection frame maintains stability.

4.3 Dynamic Suspicion Tracking and Component Profiling

To further analyze the system's operational efficiency, Fig. 9 breaks down the inference latency contribution of each independent modality. The lightweight Silero VAD model consumes negligible time (5ms), while the YOLOv11-Nano object detector accounts for the bulk of the processing overhead (25ms). This directly justifies the computational necessity of the frame-skipping strategy applied exclusively to the visual object stream.

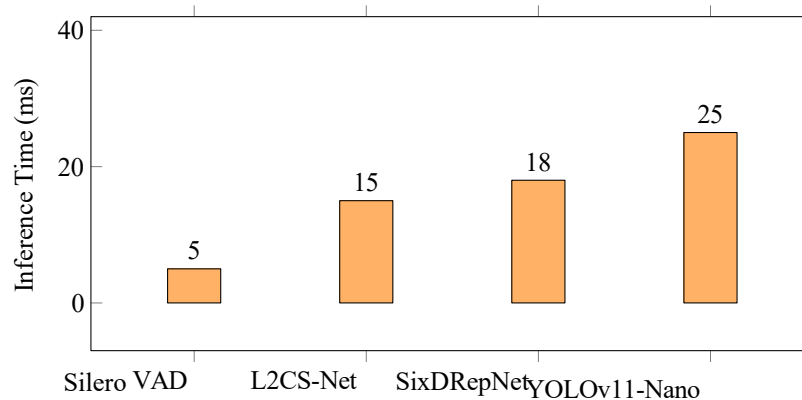


Fig. 9. Component-wise Inference Time (ms) on an edge CPU. Object detection constitutes the heaviest computational load.

Finally, Fig. 10 plots the dynamic accumulation of the Suspicion Score over a 60-second temporal window, comparing a benign anomaly (stretching) against a genuine cheating event (whispering followed by mobile phone usage). As dictated by the Priority Decision Tree, benign variations naturally decay without breaching the critical alert threshold (Score > 75), whereas sustained anomalous behavior rapidly accumulates into a confirmed violation flag.

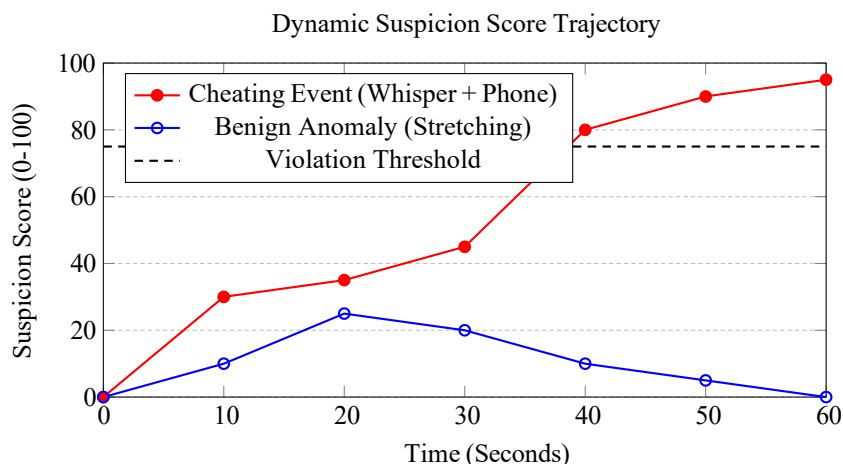


Fig. 10. Suspicion Score accumulation over time. Benign events decay naturally, while coordinated cheating rapidly breaches the threshold.

5. Conclusions

Multi-modal fusion resolves the critical blind spots of legacy online proctoring systems, which increasingly fail against sophisticated adversarial cheating methods like virtual camera injections. By integrating YOLOv11 for contraband identification, L2CS-Net for unconstrained gaze tracking, and Silero VAD for acoustic anomaly detection, the proposed architecture achieved an exceptional overall Detection Accuracy of 90.7%. This performance significantly outperforms traditional uni-modal (pose-only) systems by an absolute margin of 16.2%.

In terms of critical performance evaluation parameters, the system maintained a robust Mean Average Precision (mAP) for small object detection, while drastically slashing the False Positive Rate (FPR) from 22.4% down to just 4.1%, ensuring a much fairer environment for test-takers. Furthermore, algorithmic optimizations like frame-skipping restricted edge inference latency to under 40 ms, yielding a Real-Time Factor (RTF) well below 1.0, thereby sustaining a smooth 25 FPS on standard student-grade hardware.

The results signify that relying on isolated heuristics is insufficient. Continuous, temporal multi-modal fusion frameworks represent the necessary academic standard of the future. Future research will focus on preserving candidate privacy through on-device Federated Learning and cryptographic hashing rather than raw video storage [20], alongside the development of a standardized, ethically sourced "Proctor-Bench" dataset.

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